

Anshika Shukla

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PROFESSIONAL SUMMARY

ECE undergraduate specializing in AI/ML, Edge AI, and full-stack engineering with published research in Brain-Computer Interfaces. Builds end-to-end systems spanning embedded ML on Raspberry Pi, real-time computer vision, LLM backends, and React frontends. National hackathon finalist, GSSoC contributor, and government-recognized innovator.

EDUCATION

Banasthali Vidyapith — B.Tech, Electronics & Communication Engineering CGPA: 7.25/10 July 2023 – May 2027

- Published Research ([SSRN ID: 6763558](#)): "Advancements in Brain-Computer Interfaces for Communication Disorders and Assistive Technologies"
- Poster Presentation — WACIIS 2025 International Conference

KEY ACHIEVEMENTS

- 2nd Runner-Up — HACKVERSE (National, 30-hr, 100+ teams): Built SafeGaze achieving 95% detection accuracy under live competition conditions.
- Top 10 — HACKCELESTIA (National): Architected SatLink edge-AI platform with satellite relay delivering alerts in <10s in zero-connectivity zones.
- GSSoC 2026 Contributor (Girl Script Summer of Code) | Selected — Sanchar Saathi Initiative, Govt. of India (DoT): Recognized nationally for digital safety innovation.

EXPERIENCE

Technical Workshop Lead / Core Member — SAMARTHYA Electronics Club Sept 2024 – Present | Banasthali Vidyapith

- Boosted project completion by 40% and lab participation by 35% by designing 5+ IoT/embedded workshops for 200+ students; mentored 20+ juniors in circuit design at national hackathons.

PROJECTS

Flight-Sense-AI — Real-Time Flight Anomaly Detection | Python | Isolation Forest | OpenSky ADS-B | React | Supabase | WebSockets May 2026

- Engineered Isolation Forest anomaly detection on live ADS-B RF telemetry tracking 1,000+ aircraft over India (<200ms latency); built React + Supabase dashboard with real-time WebSocket ingestion.

SHL Assessment Recommender System | FastAPI | FAISS | Sentence Transformers | Groq LLMs | RAG May 2026

- Engineered AI recommendation engine using FAISS vector search and Sentence Transformers with Groq LLM reasoning; achieved top-3 match accuracy across 100+ SHL assessments with <1s response time.

SMART-GUARD — IoT Smart Helmet | Arduino Uno | C++ | MPU6050 | GSM | Twilio API Apr 2026

- Reduced emergency response time to <2 sec by engineering 4-sensor fusion (MPU6050, IR, MQ3, UV) accident detection with GSM/Twilio SMS+GPS alerts in C++.

AI Customer Support Agent | Python | FastAPI | Hugging Face | RAG | Prompt Engineering | Streamlit Apr 2026

- Cut manual escalation by 30% and raised first-contact resolution by 25% via RAG-powered LLM agent with 85%+ intent accuracy, <500ms latency, 100+ req/sec.

SatLink — Edge-AI Emergency Response Platform | Python | TensorFlow Lite | YOLO | OpenCV | Fourier Analysis | Twilio API | Flask Jan 2026

- Achieved 90%+ fire/flood classification via Fourier and Wavelet feature extraction on real-time camera feeds; SOS pipeline with GPS + satellite relay delivering sub-10s Twilio SMS/Call alerts in low-connectivity zones.

Real-Time Satellite Safety & Health Dashboard | YOLO-Tiny | Python | MQTT | WebSockets | Edge AI Oct 2025

- Reduced manual monitoring by 60% via YOLO-Tiny at 15+ FPS / 95%+ accuracy; MQTT/WebSocket telemetry stack at 1,000+ updates/sec, 180ms latency on edge hardware.

SignEase — Real-Time ASL Recognition | Python | TensorFlow Lite | OpenCV | Raspberry Pi | CNN | gTTS Sept 2025

- Compressed CNN 81% (45 MB to 8.5 MB) with 94% accuracy at <100ms on Raspberry Pi; 80K+ training images with 18% robustness gain via augmentation.

SafeGaze — AI Content Moderation | Python | Flask | DeepFace | Custom Deep Learning Model Mar 2025

- Cut false positives by 40% across 10K+ images; 500+ items/min at <300ms integrating DeepFace with custom NSFW model at 95% accuracy.

TECHNICAL SKILLS

Core Expertise: Edge AI · Computer Vision · Embedded ML · Deep Learning · CNNs · Signal Processing · Model Quantization · Inference Optimization · IoT Systems

AI/ML & LLMs: TensorFlow · TensorFlow Lite · Keras · Scikit-learn · OpenCV · Hugging Face · YOLO · FAISS · Sentence Transformers · RAG · Prompt Engineering

Frontend & Backend: React · FastAPI · Flask · REST APIs · WebSockets · MQTT · Supabase

Programming Languages: Python · C/C++ · JavaScript · HTML5/CSS3 · MATLAB · SQL

Hardware & Embedded: Raspberry Pi · Arduino Uno · Microcontrollers · MPU6050 · GSM/GPS Modules · Embedded Sensors

CS Fundamentals: Data Structures & Algorithms (DSA) · Object-Oriented Programming (OOP) · Database Management Systems (DBMS) · Computer Networks · Linux/Unix CLI

APIs & Tools: Groq API · Twilio API · Google Cloud APIs · Git · GitHub · VS Code · Jupyter · Streamlit · Vercel · Canva

CERTIFICATIONS & LANGUAGES

Building Data & AI Solutions on Google Cloud — 365 Data Science (100%, Mar 2026) | NASA ARSET | MATLAB Onramp — MathWorks | Generative AI — Microsoft & LinkedIn | Drone Certification — Banasthali Vidyapith | French B1 Diploma

Spoken Languages: English (Proficient) · Hindi (Native) · French (B1)